Downlink Interference Analysis of Large-Scale Non-Geostationary Orbit Constellation Based on Fibonacci Grid Method

Yunfan Zhang, Feihuang Chu, Luliang Jia, Ying Wan, Wenting Cao, and Xiaoru Chang

***Abstract*—With the increasing number of satellites, the downlink interference between non-geostationary orbit (NGSO) constellation systems is becoming more prominent. Interference-to-noise ratio (I/N) is an important evaluation index to measure the level of mutual interference, and its probability distribution is usually obtained by extrapolating the satellite orbit position and counting the time proportion of different I/N values. However, the traditional propagation method has the problems of long simulation time and low calculation efficiency. This paper proposes a new method to calculate the I/N of large-scale non-geostationary orbit constellation. By modeling large-scale constellation as uniformly distributed in space, Fibonacci grid method is used to calculate satellite position coordinate. In addition, equivalent number of satellites is defined to take into account the number of visible satellites at different latitudes. The expression of I/N is derived, and the probability distribution of I/N is obtained. On this basis, the effect of parameters of the constellations and the latitude of Earth Station on the I/N distribution is analyzed. The numerical results of the proposed method and the orbit propagation method is similar, and the calculation speed of the former is obviously faster than that of the latter.**

***Index Terms*—NGSO, downlink, interference to noise ratio, co-frequency interference analysis, Fibonacci grid method.**

I. Introduction

RECENTLY, with the proposal of 6G network, companies have deployed large-scale NGSO constellation system to provide high-speed broadband internet services [1]. According to the Federal Communications Commission (FCC) declaration data, OneWeb plans to adjust the number of satellites from 720 to 716 in phase I and expands from 716 satellites to 47,844 satellites in the second phase [2], SpaceX plans to deploy nearly 42,000 satellites [2], [3]. With the increasing number of satellites, the problem of inter-satellite co-frequency interference has become more prominent [4], [5].

There has been a lot of research on interference analysis between NGSO and GSO [6], [7], [8], [9], [10], [11], [12]. Adapting to the high variation environment and the large scale, the research on the interference between NGSO is still in its infancy [13], [14], [15]. The commonly used indexes to evaluation the interference between constellations of NGSO communications are interference-to-noise ratio (I/N), carrier-to-interference ratio (C/I), equivalent power flux density (EPFD), etc. The traditional method is to obtain the probability distribution by extrapolating the satellite orbital position and calculating the proportion of the occurrence time of different interference values. However, with the increasing number of satellites, the interference combinations increase dramatically, and the simulation of satellite orbital cycles requires long time snapshots and large computational totals, which are difficult to be effectively simulated by ordinary computers. Therefore, how to adopt effective approximate model to reduce computational complexity is an urgent problem to be solved.

In order to reduce the computational complexity, Fortes suggested that the interference analysis was accomplished by numerical methods [16], [17]. By deriving the satellite probability density function, the probability distribution of interference was calculated in the global integration, and adaptive step size and Gaussian product rule were used to reduce the computational workload, but no specific analysis steps were provided. In [18] and [19] established the system model of downlink interference of Walker constellation, and proposed a method to calculate the interference probability distribution of giant constellation based on reference satellite, but it still need a few time when the number of interfering constellation is huge. In [20], [21], [22], [23], [25], [26] proposed the idea of using stochastic geometry to ananlyse the communication performance of large-scale constellations. In [20] modeled the large-scale constellation as binomial point process (BPP) where satellites are assumed uniformly distributed in space, [23] defined the effective satellite number to take acount into the non-uniform distribution of satellites across different latitudes by modeling the constellation as nonhomogeneous poisson point process. Both methods verified the correctness

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Yunfan Zhang, Ying Wan, and Wenting Cao are with the Space Engineering Univercity, Beijing 101416, China (e-mail: z13856089081@163.com; 1400176958@qq.com; caowt6753@163.com).

Feihuang Chu is with the National Key Lab of Space Target Awareness, School of Space Information, Space Engineering University, Beijing 101416, China (e-mail: fhchu@126.com).

Luliang Jia is with the National Key Lab of Space Target Awareness, School of Space Information, Space Engineering University, Beijing 101416, China, and also with the Key Laboratory of Intelligent Space TTC&O, Ministry of Education, Space Engineering University, Beijing 101416, China (e-mail: jiallts@163.com).

Xiaoru Chang is with the Department of Electrical and Optical Engineering, Space Engineering University, Beijing 101416, China (e-mail: 604843304@qq.com).

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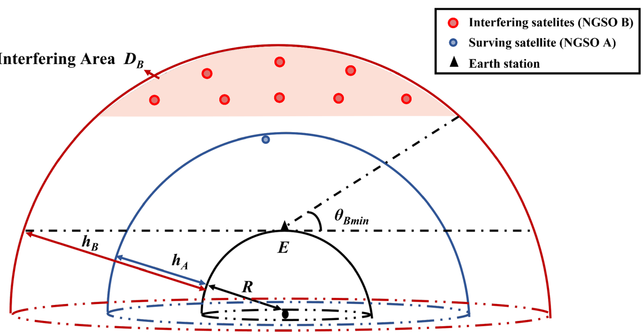


Fig. 1. NGSO B coexist with NGSO A at the same frequency, satellites in D_B cause interference.

of the simulation. However, there is a lack of research on the analysis of interference with directional antenna in LEO constellations by using stochastic geometry [24]. In addition, the mathematics of stochastic geometry is complex with many integral terms, which is computationally unfriendly. Especially, if the interference analysis of multi-layer LEO constellations is considered, the complexity brought by the stochastic geometry method will be higher.

Similar to [20], large-scale non-geostationary orbit constellation is modeled as uniformly distributed in this paper. Fibonacci grid method is used instead of the stochastic geometry to calculate the satellite position coordinates. It should be stated that orbital perturbations, collision avoidance, and inclined elliptical orbits are not considered in this paper. In addition, equivalent number of satellites is defined to take into account the number of visible satellites at different latitudes in the real scenario. Compared to orbital propagation method, the amount of computation is greatly reduced. The main contributions of this paper are as follows:

- In the author's limited knowledge, Fibonacci grid method is proposed for the first time to calculate the satellite position coordinates of large-scale non-geostationary orbit constellation, which greatly reduce the amount of computation, the numerical results of the proposed method and the orbit propagation method is similar;
- Equivalent number of satellites is defined to take into account the number of visible satellites at different latitudes, which make the proposed method better meet the reality;
- The effect of the altitude of constellation on I/N distribution is analyzed through Fibonacci grid method and the orbit propagation method.

The rest of the paper is organized as follows. Section II describes the system model. In Section III, Fibonacci grid method is introduced and downlink I/N expression is derived; Simulation analysis show that the proposed method can achieve superior performance in Section IV. Section V concludes this paper.

II. System Model

A. Downlink Interfering Scenario

As shown in Fig. 1, assuming the Earth Station E in a certain area, NGSO constellation A is the communication constellation of E, and NGSO constellation B coexist with NGSO constellation A at the same frequency (in the subsequent discussion, NGSO constellation A is denoted as NGSO A, and NGSO constellation B is denoted as NGSO B). The interfered system consists of E and one satellite in NGSO A, which is called surving satellite. The interfering system consists of several satellites in NGSO B, which are called interfering satellites. Satellites in NGSO B with an elevation greater than $\theta_{B\min}$ cause interference, the corresponding airspace is called interfering area, denoted as D_B. The Earth radius is R. The number of satellites in NGSO A is N_A, the corresponding altitude and inclination are h_A and i_A respectively. The number of satellites in NGSO B is N_B, the corresponding altitude and inclination are h_B and i_B respectively. The notation followed in this paper is summarized in Table I.

B. Propagation Model

In this paper, the shadowed-Rician (SR) fading is not considered.

1) Free Space Loss: The free space loss (FSL) is written as $L_d = [\lambda/(4\pi d)]^2$ with transmission distance d and wavelength λ.

2) Transmit Antenna Gain of Satellite: According to ITU-R S.1528 [27], the gain of satellite transmitting antenna is

$$G_1(\theta_1) = \begin{cases} G_{s\max} & \theta_1 < \phi_b, \\ G_{s\max} - 3\left(\frac{\theta_1}{\phi_b}\right)^2 & \phi_b < \theta_1 < Y, \\ G_{s\max} + L_s - 25\log\left(\frac{\theta_1}{Y}\right) & Y < \theta_1 < Z, \\ L_F & Z < \theta_1 < 180^\circ, \end{cases} \tag{1}$$

where $G_{s\max}$ is the maximum gain of satellite transmit antenna, θ_1 is the off-axis angle of satellite antenna. For LEO, we have

$$L_s = -6.75, Y = 1.5\phi_b, Z = Y(10)^{0.04(G_{s\max}+L_s-\ L_F)}. \tag{2}$$

3) Receive Antenna Gain at the Earth Station: For the Earth Station, according to the NGSO Earth Station antenna direction diagram given by ITU-R S.465 [28], the calculation formula is as follows:

$$G_2(\theta_2) = \begin{cases} G_{d\max} & \theta_2 < \phi_{\min}, \\ 32 - 25\log(\theta_2) & \phi_{\min} \le \theta_2 < 48^\circ, \\ -10 & \text{else}, \end{cases} \tag{3}$$

where $G_{d\max}$ is the maximum gain of Earth Station receive antenna, θ_2 is the off-axis angle of Earth Station antenna. According to [28], we have

$$\phi_{\min} = \begin{cases} \max(1^\circ, 100\lambda/D) & D/\lambda \ge 50, \\ \max(2^\circ, 144D/\lambda^{-1.09}) & \text{else}, \end{cases} \tag{4}$$

where D denotes the circular equivalent diameter of antenna.

C. Performance Evaluation Index

There are several evaluation indexes for satellite communication interference, which are interference-to-noise ratio (I/N), carrier-to-interference ratio (C/I) and equivalent power flux

TABLE I
SUMMARY OF MATHEMATICAL NOTATION

Notation	Description
R; h_A; h_B	Earth radius; Altitude of NGSO A; Altitude of NGSO B
i_A; i_B; i	Inclination of NGSO A; Inclination of NGSO B; Inclination of arbitrarily constellation
N_A; N_B	Total mumber of NGSO A; Total mumber of NGSO B;
N_B^{Evis}; N_B^E	The number of visible satellites in D_B at E; The equivalent number of satellites in NGSO B at E
G_1; G_2	The transmission gain; The receiving gain
θ_1; θ_{1n}	The off-axis angle of arbitrarily satellite antenna; The off-axis angle of the n-th interfering satellite antenna
θ_2; θ_{2n}	The off-axis angle of Earth Station antenna
d; d_n	Transmission distance; Transmission distance from the n-th interfering satellite to the Earth Station
P_E	The equivalent transmission power of the overlapping bandwidth
F	The overlap bandwidth factor
B_A; B_B	The communication bandwidth of NGSO A; The communication bandwidth of NGSO B
f_A; f_B	The central frequency of B_A; The central frequency of B_B
D_B; D_S; A_S	Interfering area; A certain area; The spherical area of D_S
θ_{Bmin}	The minimum elevation angle of interfering area
z_{Bmin}	The minimum altitude of Interfering area
p; p_B	The probability of a satellite appearing in a certain area; The probability of a satellite appearing in the interfering area
$\Delta\theta_\varepsilon$; $\Delta\theta_\beta$	The geocentric angle of length of D_S; The geocentric angle of width of D_S
L; L_B	The latitude of the center of D_S; The latitude of Earth Station
A_t; B_n	The surving satellite at time t; The n-th interfering satellite

density (EPFD), etc. In this paper, I/N is used to evaluate the interference level.

The interference evaluation index I/N for a single satellite communication system is defined as

$$\frac{I}{N} = \frac{P_E G_1(\theta_1) G_2(\theta_2)}{KTB} \left(\frac{\lambda}{4\pi d}\right)^2 \tag{5}$$

where P_E is the equivalent transmission power of the overlapping bandwidth, θ_1 is the off-axis angle of satellite antenna, $G_1(\theta_1)$ is the transmission gain, θ_2 is the off-axis angle of Earth Station antenna, $G_2(\theta_2)$ is the receiving gain, K is the Boltzmann constant, T is the noise temperature at the receiving end of the Earth Station, B is the satellite communication bandwidth, d is transmission distance and λ is wavelength.

III. Establishment of Downlink I/N Expression

It can be seen from (5) that I/N is determined by θ_1, θ_2 and d when the satellite transmitter power, bandwidth, gain, and other characteristics are given. θ_1, θ_2 and d are determined by the position of the Earth Station and the satellites. The surving satellite position can be got by orbit propagation, so we focus on calculation of the interfering satellite position. NGSO B is assumed as a large-scale constellation and uniformly distributed in space. Fibonacci grid method is introduced in section A to calcuelate the interfering satellite position.

A. Fibonacci Grid Method

Fibonacci grid method is a mathematical method used to solve the problem of evenly distributing multiple points on a sphere.

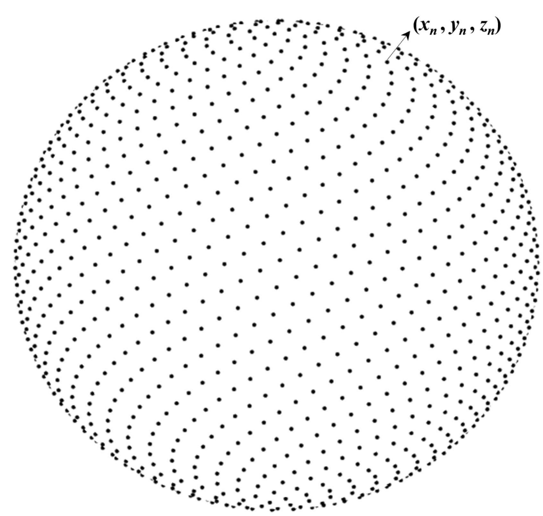


Fig. 2. Fibonacci grid point on the unit sphere.

The historical origin of Fibonacci grid method is discussed in [29], and the method of how to evenly distribute N points on a unit sphere is given in [30]. As shown in Fig. 2, a rectangular coordinate system is established with the center of the sphere as the origin. On the unit sphere, the coordinates of N evenly distributed points are as follows:

$$\begin{cases} x_n = \sqrt{1 - z_n^2}\cos(2\pi n\phi), \\ y_n = \sqrt{1 - z_n^2}\sin(2\pi n\phi), \\ z_n = \frac{2n-1}{N} - 1, \end{cases} \quad n = 1, 2, \ldots, N. \tag{6}$$

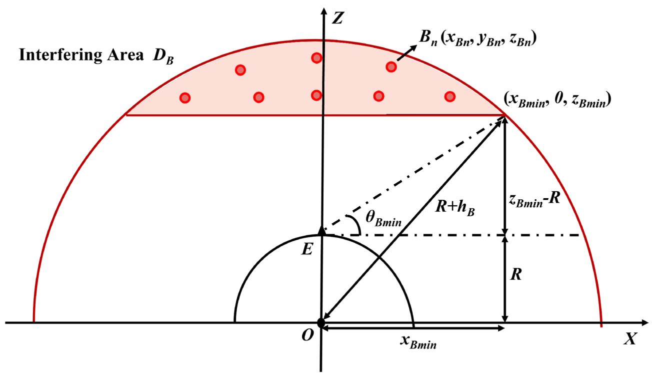


Fig. 3. Projection on the XOZ plane. Calculate $z_{B\min}$ from $\theta_{B\min}$.

The constant $\phi = (\sqrt{5} - 1)/2 \approx 0.618$ is the golden section point.

According to (6), the coordinates of each point are uniquely determined by the serial number n, and the altitude of coordinate points monotonically increases with the increase of n.

B. Interfering Satellite Position Coordinate

Using the Fibonacci grid method, the coordinates of all the satellites in NGSO B can easily got as

$$B_n = (x_{Bn}, y_{Bn}, z_{Bn}), \ \ n = 1, 2, \ldots, N_B, \tag{7}$$

where

$$\begin{cases} x_{Bn} = (R + h_B)\sqrt{1 - z_n^2}\cos(2\pi n\phi), \\ y_{Bn} = (R + h_B)\sqrt{1 - z_n^2}\sin(2\pi n\phi), \quad n = 1, 2, \ldots, N_B. \\ z_{Bn} = (R + h_B)\left(\frac{2n-1}{N_B} - 1\right), \end{cases} \tag{8}$$

n is the sequence number of the satellite, the corresponding satellite is denoted as the n-th interfering satellite. According to (6), the coordinates of the satellite are uniquely determined by n, and the altitude of the satellite increases monotonically with n.

Lemma 1: The coordinates of the interfering satellites in the interfering area D_B are

$$B_n = (x_{Bn}, y_{Bn}, z_{Bn}), n = \left\lfloor \frac{N_B\left(\frac{z_{B\min}}{R+h_B} + 1\right) + 1}{2} \right\rfloor, \ldots, N_B, \tag{9}$$

where

$$z_{B\min} = \cos^2\theta_{B\min}\left(R + tan\theta_{B\min}\sqrt{\frac{(R + h_B)^2}{\cos^2\theta_{B\min}} - R^2}\right).$$

Proof: With the center of the earth as the origin and the line between the center and the earth station as the Z-axis, the space rectangular coordinate system is established.

Since only satellites in D_B can cause interference, the coordinates of satellites in D_B are the ones needed to calculate. In order to facilitate calculation, the space is project onto the XOZ plane as shown in Fig. 3. The minimum value of the inner altitude is denoted as $z_{B\min}$, and if the Z-axis coordinate of the satellite is higher than $z_{B\min}$, the corresponding satellite appears in D_B. It is easy to get the equation

$$\begin{cases} {x_{B\min}}^2 + {z_{B\min}}^2 = (R + h_B)^2, \\ \frac{z_{B\min} - R}{x_{B\min}} = tan\theta_{B\min}, \end{cases} \tag{10}$$

we have

$$z_{B\min} = \cos^2\theta_{B\min}\left(R + \tan\theta_{B\min}\sqrt{\frac{(R + h_B)^2}{\cos^2\theta_{B\min}} - R^2}\right), \tag{11}$$

let

$$z_n \geq z_{B\min}, \tag{12}$$

we get

$$n \geq \frac{N_B\left(\frac{z_{B\min}}{R+h_B} + 1\right) + 1}{2}. \tag{13}$$

The coordinates of the interfering satellite in the interfering area can be derived as

$$B_n = (x_{Bn}, y_{Bn}, z_{Bn}), n = \left\lfloor \frac{N_B\left(\frac{z_{B\min}}{R+h_B} + 1\right) + 1}{2} \right\rfloor, \ldots, N_B. \tag{14}$$

□

C. Equivalent Number of Satellites

In this section, the number of visible satellites in D_B at E is discusssed, which is denoted as N_B^{Evis}. NGSO B is modeled as uniformly distributed in Section B, so that N_B^{Evis} is equal wherever E is positioned. However, in the real scenario, N_B^{Evis} varies with the latitude of E. To calculate N_B^{Evis}, we refer to the method in [31].

The expression for the calculation of the probability of appearance of a satellite in a spatial region of the sphere is given in [31]. For a certain airspace D_S, the probability of occurrence of one satellite of the constellation is

$$p = \frac{A_S}{2\pi^2 \sin\left(arccos\frac{\cos i}{\cos L}\right)\cos L}, \tag{15}$$

$$A_S = \frac{\pi}{4}\Delta\theta_\varepsilon\Delta\theta_\beta(rectangle), \tag{16}$$

where A_S is the spherical area of the airspace D_S, i is the inclination, L is the latitude of the center of the calculated airspace D_S, $\Delta\theta_\varepsilon$ and $\Delta\theta_\beta$ are the geocentric angle of length and width of D_S. Based on the (16) and (17), the following lemma is obtained.

Lemma 2: If the latitude of E is L_B, the number of visible satellites in D_B at E is

$$N_B^{Evis} = \left\lfloor N_B \times \frac{arccot^2\left(\frac{z_{B\min} - R}{z_{B\min}}\tan\theta_{B\min}\right)}{2\pi\sin\left(arccos\frac{\cos i_B}{\cos L_B}\right)\cos L_B} \right\rfloor, \tag{17}$$

where $z_{B\min}$ can be got from (12).

Proof: The probability of a satellite appearing in D_B is denoted as p_B. As shown in the Fig. 4, $\Delta\theta_\varepsilon = \Delta\theta_\beta$ since D_B

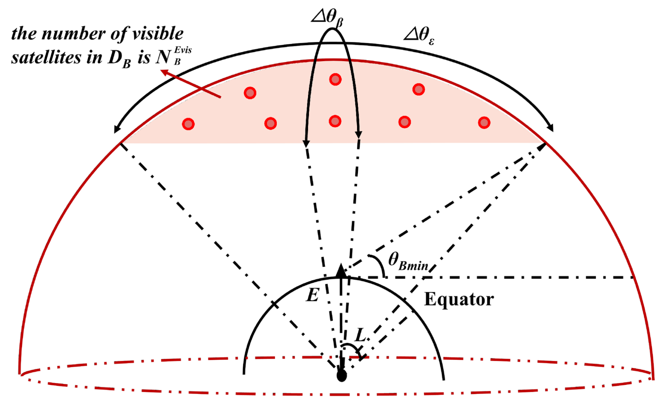


Fig. 4. $\Delta\theta_\varepsilon = \Delta\theta_\beta$ since D_B is a spherical cap. It should be noted that the radius of Earth is actually large, D_B is relatively small.

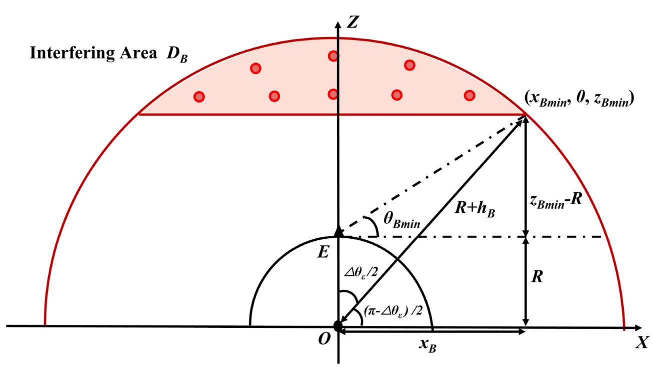


Fig. 5. Projection on the XOZ plane. Calculate $\Delta\theta_\varepsilon$ from $\theta_{B\min}$.

is the spherical cap. In addition, it is easily got that $L = L_B$, $i = i_B$. Notice the two right triangles in Fig. 5, according to the definition of the tangent function, we have

$$\begin{cases} \tan\theta_{B\min} &= \frac{z_{B\min}-R}{x_B}, \\ \tan\frac{\pi-\Delta\theta_\varepsilon}{2} &= \frac{z_{B\min}}{x_B}, \end{cases} \tag{18}$$

so that

$$\Delta\theta_\varepsilon = 2arccot\left(\frac{z_{B\min}-R}{z_{B\min}}\tan\theta_{B\min}\right). \tag{19}$$

Substitute (19) to (15), we have

$$p_B = \frac{A_S}{2\pi^2\sin\left(arccos\frac{\cos i}{\cos L}\right)\cos L}$$

$$= \frac{\frac{\pi}{4}2arccot^2\left(\frac{z_{B\min}-R}{z_{B\min}}\tan\theta_{B\min}\right)}{2\pi^2 sin\left(arccos\frac{\cos i_B}{\cos L_B}\right)\cos L_B}$$

$$= \frac{arccot^2\left(\frac{z_{B\min}-R}{z_{B\min}}\tan\theta_{B\min}\right)}{2\pi\sin\left(arccos\frac{\cos i_B}{\cos L_B}\right)\cos L_B}, \tag{20}$$

the number of visible satellites in D_B at E is $N_B^{Evis} = N_B \times p_B$, (17) is got by substituting (20) into it. □

On the other hand, since NGSO B is assumed to be uniformly distributed, the number of visible satellites in D_B at E can also be obtained as

$$N_B \times \frac{S}{4\pi(R+h_B)^2},$$

where S is the spherical area of D_B,

$$S = 2\pi(R+h_B)(R+h_B-z_{B\min}).$$

To equalize the number of visible satellites in D_B at E calculated by these two methods, we adjust the total number of satellites N_B to N_B^E corresponding to different latitudes of E, so that

$$N_B^E \times \frac{S}{4\pi(R+h_B)^2} = N_B^{Evis}. \tag{21}$$

N_B^E is denoted as the equivalent number of satellites in NGSO B at E, whose expression is given in (22) shown at the bottom of the page.

D. Downlink I/N Expression

In the previous section, the coordinates of the interfering satellites are derived by using Fibonacci grid method. In addition, the serving satellite of the Earth Station at time t is denoted as A_t (its coordinate is denoted as (x_{At}, y_{At}, z_{At})). As shown in Fig. 6, the downlink I/N expression is derived as follow.

Theorem 1: The aggregate I/N of the Earth Station with lattitude L_B at time t is

$$\left(\frac{I}{N}\right)_t = \sum_{n=\left\lceil\frac{N_B^E\left(\frac{z_{B\min}}{R+h_B}+1\right)+1}{2}\right\rceil}^{N_B^E} \frac{P_E G_1(\theta_{1n})G_2(\theta_{2n})}{KTB}\left(\frac{\lambda}{4\pi d_n}\right)^2, \tag{23}$$

where θ_{1n} is the off-axis angle of the n-th interfering satellite antenna, θ_{2n} is the off-axis angle of Earth Station antenna, d_n is the distance from the n-th interfering satellite to the Earth

$$N_B^E = \left\lfloor 2\times N_B \times p_B \times \frac{R+h_B}{R+h_B-z_{B\min}}\right\rfloor = \left\lfloor 2\times N_B \times \frac{arccot^2\left(\frac{z_{B\min}-R}{z_{B\min}}\tan\theta_{B\min}\right)^2}{2\pi\sin\left(arccos\frac{\cos i_B}{\cos L_B}\right)\cos L_B} \times \frac{R+h_B}{R+h_B-z_{B\min}}\right\rfloor$$

$$= \left\lfloor N_B \times \frac{arccot^2\left(\frac{z_{B\min}-R}{z_{B\min}}\tan\theta_{B\min}\right)^2}{\pi\sin\left(arccos\frac{\cos i_B}{\cos L_B}\right)\cos L_B} \times \frac{R+h_B}{R+h_B-z_{B\min}}\right\rfloor. \tag{22}$$

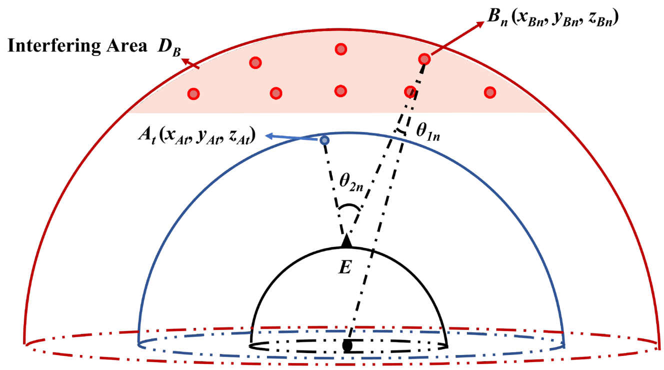


Fig. 6. At time t, the surving satellite A_t is interfered by the n-th interfering satellite B_n. The sum of all the interference is the aggregate I/N value.

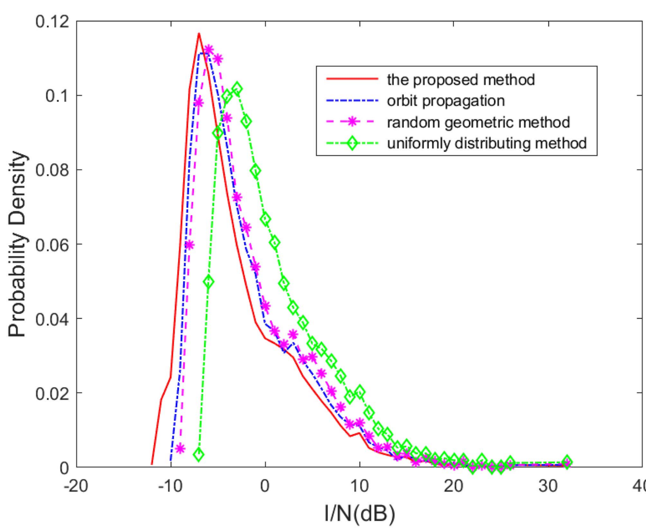


Fig. 7. Comparison of interference probability distribution between a 60-day orbit propagation, random geometric method, uniformly distributing method and the proposed method.

Station, the rest of the parameters are described in previous section.

Proof: This is almost an obvious proof strategy from (9) and (22). □

G_1, G_2 is given by (1) and (3), the expression of $P_E, \theta_{1n}, \theta_{2n}$ and d_n is given as follows:

P_E: Assuming that the communication bandwidth of NGSO A is B_A, the central frequency is f_A, the communication bandwidth of NGSO B is B_B, the central frequency is f_B, and the transmitting power of NGSO B is P_B, then it can be obtained

$$P_E = P_B \times F, \tag{24}$$

where

$$F = \frac{\left|\left(f_A - \frac{B_A}{2}, f_A + \frac{B_A}{2}\right) \cap \left(f_B - \frac{B_B}{2}, f_B + \frac{B_B}{2}\right)\right|}{B_B} \tag{25}$$

is called the overlap bandwidth factor.

θ_{1n}: Assuming that that the center direction of the transmitting antenna of the interfering satellite is pointing to the center of the Earth. According to the previous section, the coordinates of the interfering satellite are (x_{Bn}, y_{Bn}, z_{Bn}), the coordinates of the Earth Station $(0, 0, R)$, the coordinates of the center of the Earth $(0, 0, 0)$, the angle can be obtained as

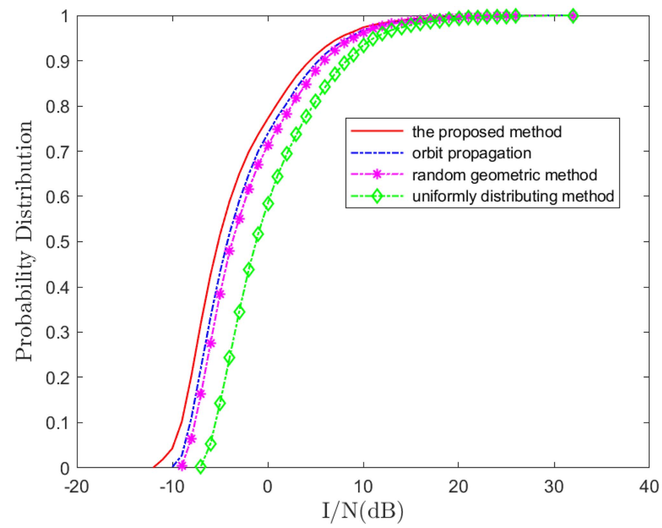


Fig. 8. Comparison of interference probability distribution between a 60-day orbit propagation, random geometric method, uniformly distributing method and the proposed method.

$$\theta_{1n} = \arccos\left(\frac{(x_{Bn}, y_{Bn}, z_{Bn}) \cdot (x_{Bn}, y_{Bn}, z_{Bn} - R)}{\|(x_{Bn}, y_{Bn}, z_{Bn})\| \, \|(x_{Bn}, y_{Bn}, z_{Bn} - R)\|}\right), \tag{26}$$

where $||\cdot||$ represents the Euclidean norm.

θ_{2n}: According to the previous section, the Earth Station coordinates $(0, 0, R)$, the coordinates of the interfering satellite are (x_{Bn}, y_{Bn}, z_{Bn}), the coordinate of the surving satellite is (x_{At}, y_{At}, z_{At}), the angle can be obtained as

$$\theta_{2n} = \arccos\left(\frac{(x_{Bn}, y_{Bn}, z_{Bn} - R) \cdot (x_{At}, y_{At}, z_{At} - R)}{\|(x_{Bn}, y_{Bn}, z_{Bn} - R)\| \, \|(x_{At}, y_{At}, z_{At} - R)\|}\right). \tag{27}$$

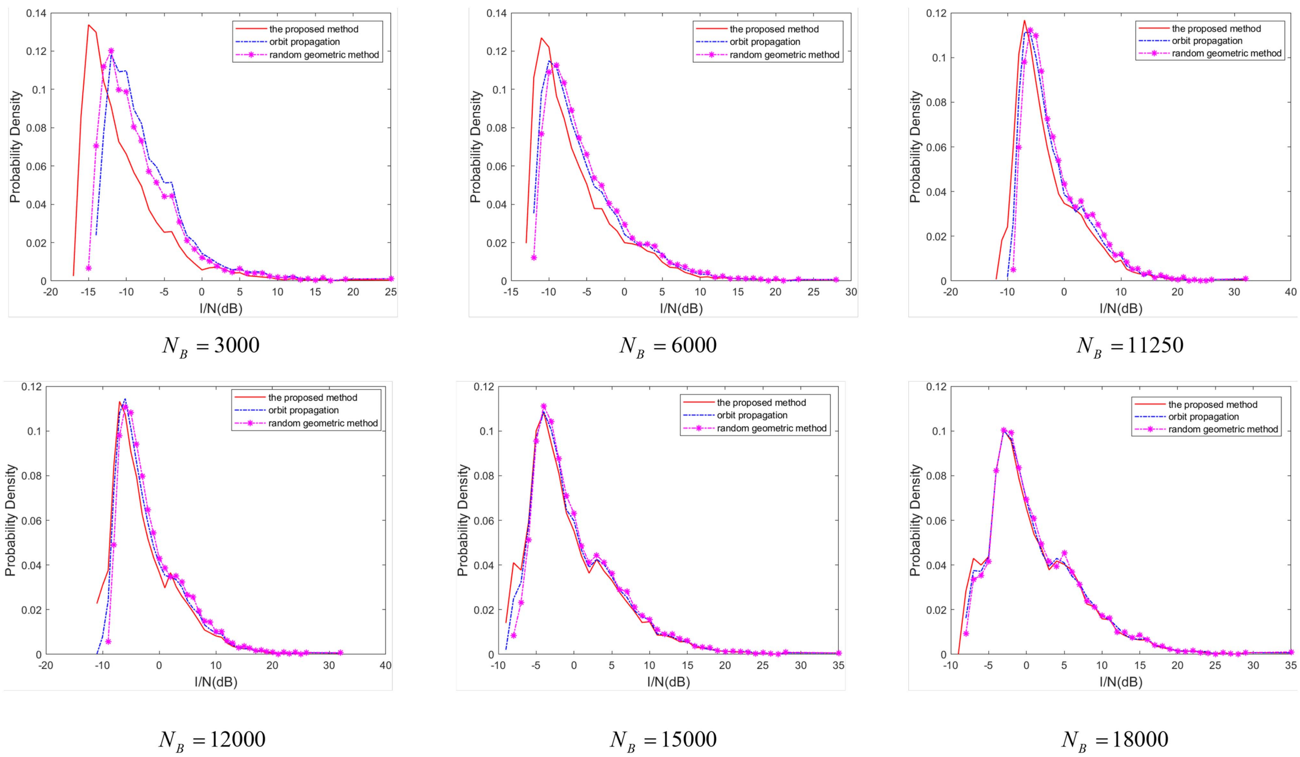
d_n: Its expression is obtained as

$$d_n = \sqrt{x_{Bn}^2 + y_{Bn}^2 + (z_{Bn} - R)^2}. \tag{28}$$

Substituting the expression of (1), (2), (22) and (24)–(28) into (23), we can get the complete mathematical expression of $(I/N)_t$. Different t (namely different serving satellite) corresponds to different I/N value. In this way, we get the I/N distribution, and the effect of the altitude of constellation on the I/N distribution can be obtained by controlling the variable method, which will be shown in the simulation analysis in Section IV.

IV. Simulation Analysis

STK and MATLAB are used to simulate the interference between two constellations. The simulation parameters are shown in Table II. NGSO A is set as a normal NGSO constellation, and NGSO B as a large NGSO constellation which is approximate uniformly distributed in space. NGSO A adopts the distance-first access scheme, and its satellite uses a dynamic spot beam with



$N_B = 3000$ $N_B = 6000$ $N_B = 11250$

$N_B = 12000$ $N_B = 15000$ $N_B = 18000$

Fig. 9. The I/N probability distribution under different N_B.

TABLE II
SIMULATION PARAMETERS OF CONSTELLATION SYSTEM

Parameter	NGSO A	NGSO B
Orbital altitude	1000 km	1200 km
Orbital inclination	88°	86°
Number of satellites	200	11250
Orbital planes	10	75
Maximum transmitting power	7 dBW	7 dBW
Downlink bandwidth	36 MHz	49 MHz
Downlink carrier frequency	20.075 GHz	20.075 GHz
Peak gain of receiving antenna	38.7 dB	39.9dB
Receiving antenna beam width	2°	2°
ES Receiver noise temperature	270K	
Longitude and latitude of Earth Station	[0° E, 69° N]	
Step size	5s	

TABLE III
THE SIMULATION DURATION

Method	Duration
60-day orbit propagation	16712 s
random geometric method	776 s
Proposed method	169 s

a staring antenna. The antenna diagram is ITU-RS.465-5 [28]. NGSO B is a constellation using fixed beams and ground-facing antennas, and the antenna pattern is ITU-RS.1528 [27].

Figs. 7 and 8 show the I/N probability distribution and cumulative probability distribution, respectively. It can be seen that the results given by the orbit propagation simulation, the random geometric method and the proposed method are close. Accurate results cannot be obtained by the uniformly distributing method, because it cannot simulate the different distribution of satellites in different dimensions. In this paper, the proposed method corresponds to the case of an infinite number of days, Table III shows the simulation time of the two methods (3.0 GHz PC, 8 GB RAM), and the proposed method is more efficient than the orbit propagation and random geometric method. This is due to the fact that the stochastic geometry approach introduces many integral terms, which leads to slower calculations.

In addition, a comparison of the proposed method with the orbit extrapolation method and the stochastic geometry method for different numbers of interfering satellites is given, as shown in Figs. 9 and 10. It can be seen that when the number of satellites is small, the error of the results of this paper's method is larger. This is due to the fact that when the number of satellites is small, its local uniformity of distribution is not obvious. And when the number of satellites is larger, the error of the results of this paper's method is getting smaller and smaller. This is due to the fact that with the increase in the number of satellites, its localized uniform distribution is better with a larger number of satellites. In contrast, the method of random geometry has better calculation results than the method of this paper, which is due to the fact that this paper is directly modeling the giant

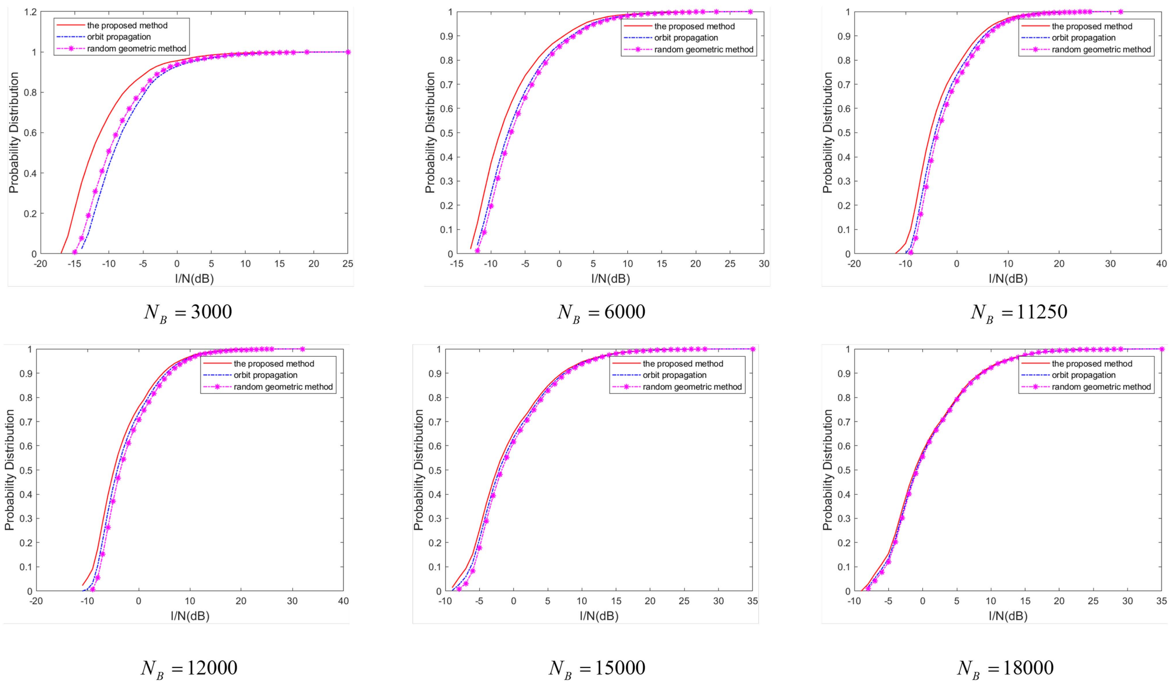


Fig. 10. The cumulative probability distribution different N_B.

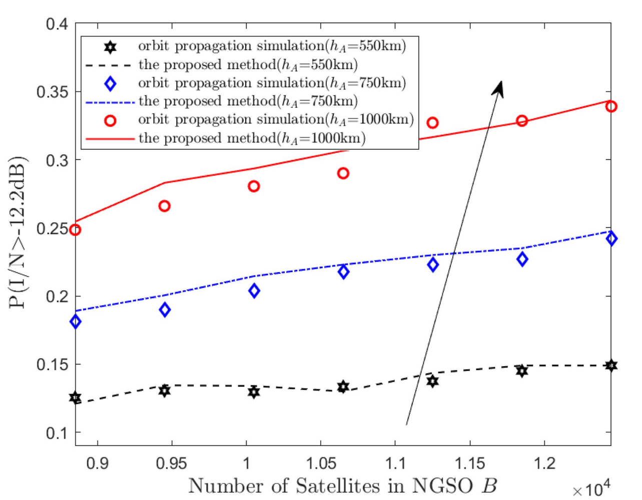


Fig. 11. Effect of different altitudes of NGSO A and different number of satellites in NGSO B on P(I/N > -12.2 dB).

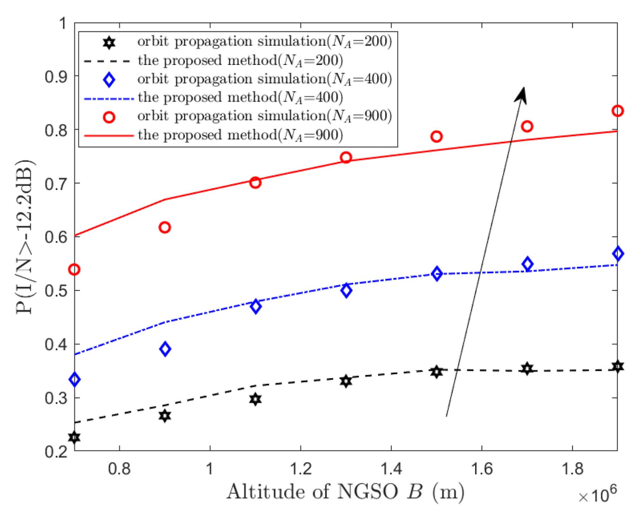


Fig. 12. Effect of different altitudes of NGSO B and different number of satellites in NGSO A on P(I/N > -12.2 dB).

interference constellation as a fixed locally uniformly distributed constellation, while the expression of random geometry according to the probability of occurrence more accurately simulates the distribution of the satellites, but due to the number of integral terms, its computational complexity is larger. Moreover, the method in this paper has a greater advantage in calculating the interference in multi-layer constellations, while it will bring more complex calculations if the stochastic geometry approach is used.

Fig. 11 shows the effect of the altitude of NGSO A and the number of satellites in NGSO B on P(I/N > -12.2 dB). It can be seen that P(I/N > -12.2 dB) increases as the number of satellites in NGSO B increases and the altitude of NGSO A increases. This is because whether the altitudes of NGSO A or the number of satellites in NGSO B increases will lead to the decrease of θ_{2n}. Then $G_2(\theta_{2n})$ increases, which leads to the increase of I/N.

Fig. 12 shows the effect of the altitude of NGSO B and the number of satellites in NGSO A on P(I/N > -12.2 dB). It can be seen that P(I/N > -12.2 dB) increases as the number of satellites in NGSO A increases and the altitude of NGSO B increases. This is because whether the altitudes of NGSO B or the number of satellites in NGSO A increases, which will lead to the decrease of θ_{1n}. So $G_1(\theta_{1n})$ increases, which leads to the increase of I/N. When the the altitude of NGSO B increases to

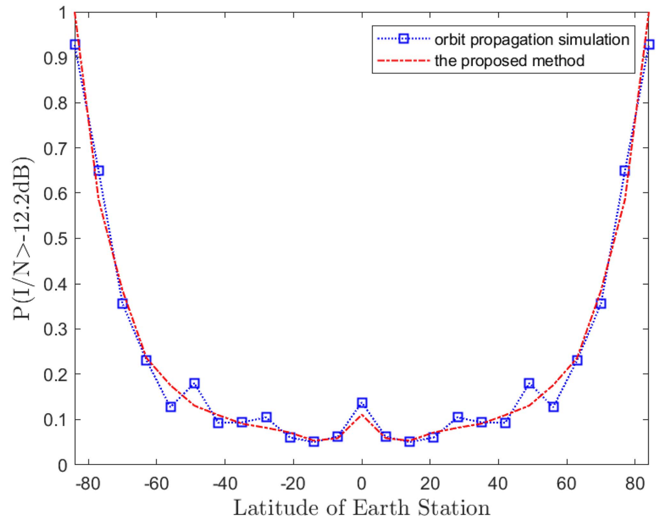


Fig. 13. Effect of different latitudes of Earth Station on P(I/N $>$ -12.2 dB).

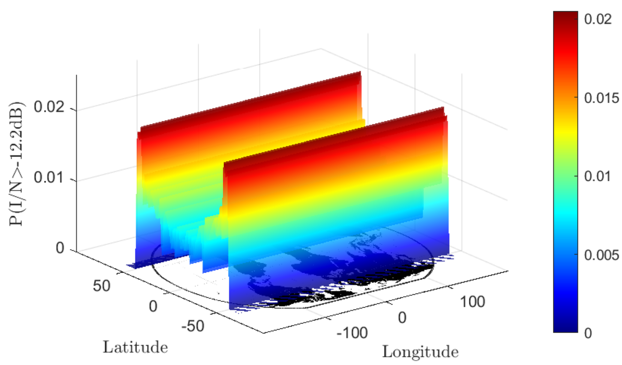


Fig. 14. Schematic of the interference scenario between SpaceX and OneWeb satellite systems. Oblique view.

very high, I/N decreases at last in our simulation. It's due to the power attenuation in the space.

Fig. 13 shows the effect of latitudes of Earth Station on P(I/N > -12.2 dB). It can be seen that P(I/N > -12.2 dB) increases as the latitudes of Earth Station increases. This is because as the latitudes of Earth Station increases, the equivalent number of satellites N_B^E increases due to (22), which leads to the increase of I/N.

We also apply this method to analyze the interference between the Starlink satellite system (with an orbital inclination of 53°) and the OneWeb satellite system (with an orbital inclination of 88°). Figs. 14 and 15 depict oblique and top views, respectively, illustrating the interference scenario between SpaceX and OneWeb satellite systems. As depicted in the figures, minimal interference is observed at low latitudes due to a lower number of visible satellites with a reduced probability of causing significant interference. However, around 50°, intense interference occurs as a result of an increased number of visible satellites leading to overlapping interferences. Selective shutdowns should be considered for certain satellites to ensure protective measures are implemented. At higher latitudes, where there is limited visibility from Earth stations, interference diminishes owing to the configuration of the Starlink constellation.

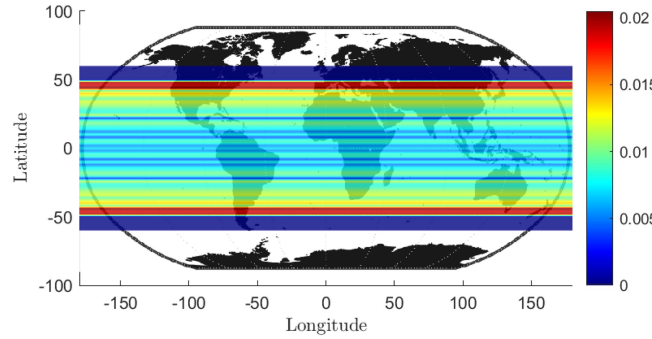


Fig. 15. Schematic of the interference scenario between SpaceX and OneWeb satellite systems. Top view.

V. Conclusion

In this paper, a new method is proposed to analyse the downlink interference of large-scale non-geostationary orbit constellation. By modeling large-scale constellation as uniformly distributed, Fibonacci grid method is used to calculate the coordinates of the satellites. In addition, the equivalent number of satellites is defined and calculated to take into account the number of visible satellites at different latitudes. The downlink I/N expression is derived, and the effect of parameters of the constellations and the latitude of Earth Station on the I/N distribution is analyzed. The numerical results of the proposed method and the orbit propagation method is similar, and the calculation speed of the former is obviously faster than that of the latter. The scalable results presented here are not only applicable to a single constellation but also to massive satellite networks that merge several different constellations.

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Yunfan Zhang received the B.S. and M.S. degrees in mathematics from Peking University, Beijing, China, in 2010 and 2014, respectively. He is currently working with Space Engineering University, Beijing, China. His research interests include learning theory, interference analysis, and random access in satellite communication.

Feihuang Chu received the M.S. and Ph.D. degrees from the Electronic Engineering Institute of PLA, Hefei, China in 1997 and 2003, respectively. He is currently working with Space Engineering University, Beijing, China. His research interests include learning theory, satellite communication, and communication anti-jamming technology.

Luliang Jia received the M.S. and Ph.D. degrees in communications and information systems from College of Communications Engineering, PLA University of Science and Technology, Nanjing, China in 2014 and 2018, respectively. He is currently an Assistant Professor with School of Space Information, Space Engineering University, Beijing, China. His research interests include game theory, learning theory, satellite communication, and communication anti-jamming technology.

Ying Wan received the M.S. degree from Space Engineering University, Beijing, China, in 2022. She is currently working with School of Space Information, Space Engineering University. Her main research interests include intelligent optimization algorithms, satellite communication, and satellite network.

Wenting Cao received the B.S. degree from Space Engineering University, Beijing, China, in 2022. where she is a Graduate Student. Her research interests include learning theory, satellite communication, and communication anti-jamming technology.

Xiaoru Chang received the M.E degree in optical engineering from the South China Normal University, Guangzhou, China, in 2015. She is currently working with Space Engeering University, Beijing, China. Her research interests include laser and optical measurement.